Proof for Diamond Graph G

Zofia Kuriata

2026-09-09

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1 Problem

We would like to prove that 0.54413127767139846114 is the value of $\mathrm{mod}(G)$ and it is achieved by a partition where four clusters have 184 diamonds and 1 path vertex, and the rest of the diamonds have their own clusters — Figure 1.

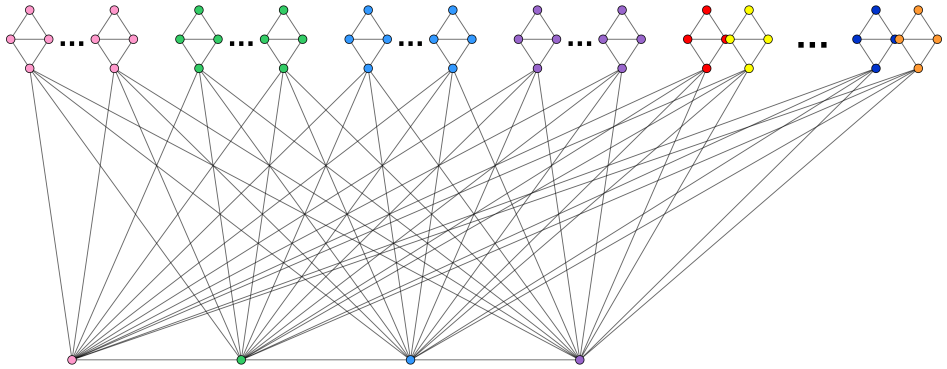


Figure 1: The best found clustering for modularity.

If we proved it, we would also prove that there indeed exists a graph G such that $\mathrm{mod}^*(G) > \mathrm{mod}(G)$, which is our *diamond graph* G.

2 Definitions and Notation

Let $G = (V, E)$ be the considered diamond graph and $m = |E| = 81003$ be the total number of edges in G.

- d_i — the i-th diamond subgraph (where $e(d_i) = 5$ and $\mathrm{vol}(d_i) = 14$),
- t_i — the i-th top triangle of diamond d_i,
- c_i — the connector vertex of i-th diamond attached to the path,
- p_1, p_2, p_3, p_4 — vertices forming the central path (path vertices),
- $\mathrm{vol}(C) = \sum_{v \in C} \deg(v)$ — the volume of cluster C (sum of degrees of vertices in C),
- $e(C)$ — the number of internal edges in cluster C,
- $\mathrm{mod}_{\mathcal{C}}(G) = \sum_{C \in \mathcal{C}} \left[\frac{e(C)}{m} - \left(\frac{\mathrm{vol}(C)}{2m} \right)^2 \right]$ — modularity of graph G under partition $\mathcal{C}$,
- $\mathcal{A}$ — partition shown in Figure 1 scoring 0.54413127767139846114,
- A_1, A_2, A_3, A_4 — clusters in $\mathcal{A}$ containing 184 diamonds and 1 path vertex each,
- $A_5, \ldots, A_{8268}$ — clusters in $\mathcal{A}$ containing exactly 1 standalone diamond.

Lemma 1. *Let $G = (V, E)$ be a graph and $\mathcal{A}$ be a partition of vertices of G such that $mod_{\mathcal{A}}(G) = mod(G)$. Then $\forall_{A \in \mathcal{A}}\, mod_A(G) \geq 0$.*

Lemma 2. *Let $G = (V, E)$ be a graph and $\mathcal{A}$ be a partition of vertices of G such that $mod_{\mathcal{A}}(G) = mod(G)$. Then every community in partition $\mathcal{A}$ induces a connected subgraph.*

3 Structural Analysis

3.1 Splitting Triangles and Diamonds

First, we will show that there is no point in splitting the top triangle in each diamond. Let a, b, c be vertices of a triangle such that $deg(c) = 2$. At first, let's take a look at the case when the vertex a (or symmetrically b) is in a different community than b and c. There are three main scenarios for such clustering.

1. a, b, c are in separate clusters.
   Let's notice that if c is grouped neither with b nor a, then it must be in its own one-vertex community (Lemma 2). Similarly for a and b - if they are not together, then at least one of them also must be alone (because there is only one remaining neighbour of them – the connector), let's assume that a is alone. Let $C_1 = \{c\}, C_2 = \{a\}, C_1' = \emptyset$ and $C_2' = \{a, c\}$. We will show that joining C_1 and C_2 will increase mod score

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{1}{m} - \frac{5^2}{4m^2} + \frac{2^2 + 3^2}{4m^2} \\
&= \frac{4m - 25 + 4 + 9}{4m^2} \\
&= \frac{4m - 12}{4m^2} > 0.
\end{aligned}$$

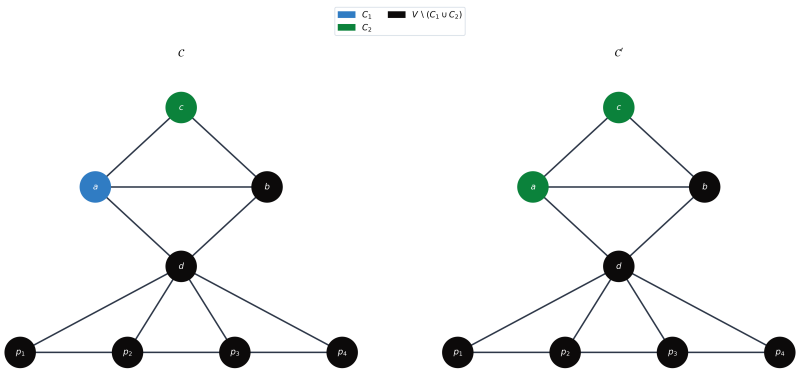


Figure 2: Merging two alone triangle vertices.

Therefore in optimal partitioning, no triangle can be split into three clusters.

2. a, c together, b in a separate cluster.
   We need to consider three scenarios, depending on where the connector is located.

   (a) Connector (d) is in $b's$ cluster.
   Let $b, d \in C_1, C_2 = \{a, c\}, C_1' = C_1 \setminus \{b\}$ and $C_2' = \{a, c, b\}$. The difference is

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{2}{m} - \frac{1}{m} - \frac{-6vol(C_1) + 9}{4m^2} - \frac{6vol(C_2) + 9}{4m^2} \\
&= \frac{4m + 6vol(C_1) - 6 \cdot 5 - 18}{4m^2} \\
&= \frac{4m + 6vol(C_1) - 48}{4m^2} > 0.
\end{aligned}$$

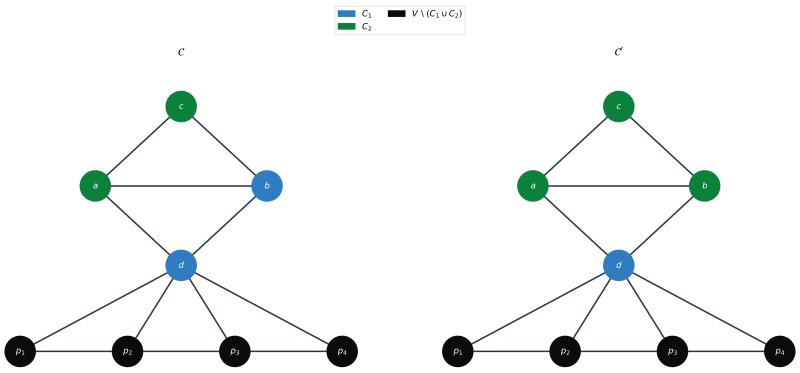


Figure 3: Moving b from the connector to the triangle.

   (b) Connector (d) is in the cluster containing a, c (b is alone).
   Let $C_1 = \{b\}, a, c, d \in C_2, C_1' = \emptyset$ and $C_2' = C_2 \cup \{b\}$. The difference is

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{3}{m} - \frac{6vol(C_2) + 9}{4m^2} + \frac{9}{4m^2} \\
&= \frac{12m - 6vol(C_2)}{4m^2} \geq 0.
\end{aligned}$$

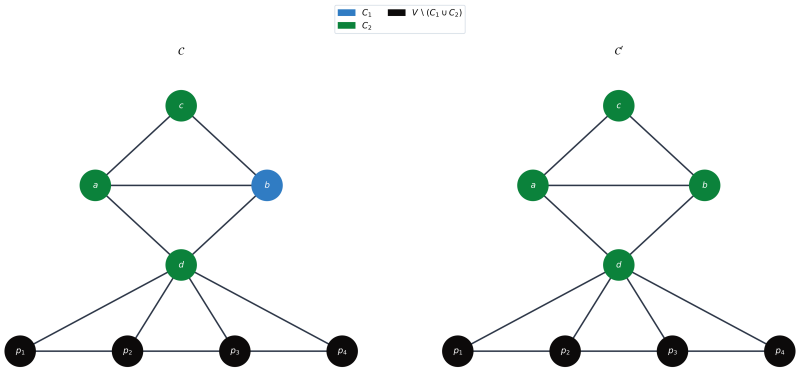


Figure 4: Moving b to the whole diamnod.

(c) Connector is in a separate cluster.
Let $C_1 = \{b\}, C_2 = \{a, c\}, C_1' = \emptyset$ and $C_2' = \{a, c, b\}$. The difference is

$$\begin{aligned} &\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\ &= \frac{2}{m} - \frac{6vol(C_2) + 9}{4m^2} + \frac{9}{4m^2} \\ &= \frac{8m - 6 \cdot 5}{4m^2} > 0. \end{aligned}$$

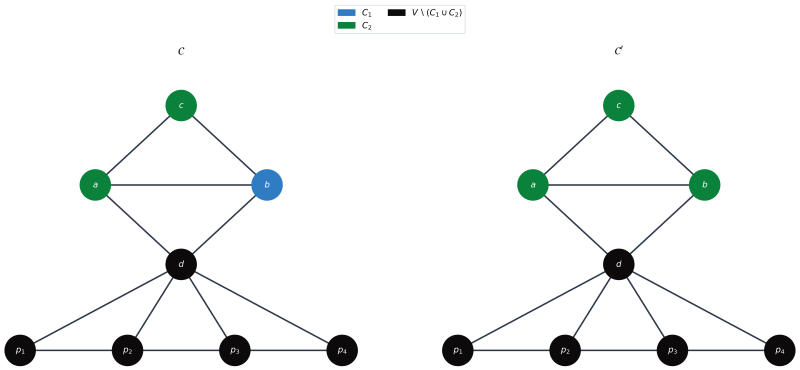


Figure 5: Moving b from its own cluster to the rest of the triangle.

Therefore, we can see that neither a nor b can be separated from the rest of its triangle. The last thing we need to show is that c also cannot be separated.

3. a, b together, c in its own one-vertex cluster.
Let $C_1 = \{c\}, a, b \in C_2, C_1' = \emptyset$ and $C_2' = C_2 \cup \{c\}$. The difference is

$$\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G)$$

$$= \frac{2}{m} - \frac{4vol(C_2) + 4}{4m^2} + \frac{4}{4m^2}$$
$$= \frac{2m - vol(C_2)}{4m^2} > 0.$$

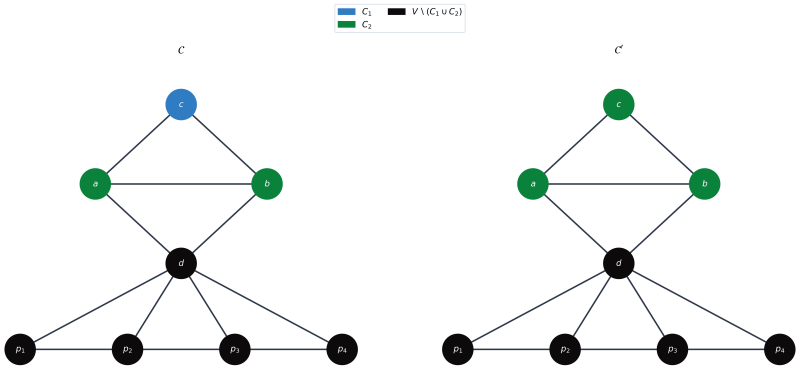


Figure 6: Moving c to the rest of the triangle.

So, in optimal partitioning no triangle is split.

The second question is: should we join the triangles with other vertices (excluding the connector)?

Let $\mathcal{C}$ and $\mathcal{C}'$ be clusterings such that $C_1, C_2 \in \mathcal{C}$ and $C_1', C_2' \in \mathcal{C}'$, where $C_1 = \{t_1\}$, $c_1 \notin C_2$, $C_1' = \emptyset$, and $C_2' = C_2 \cup \{t_1\}$.

Consider the difference

$$\begin{aligned}
&\text{mod}_{\mathcal{C}'}(G) - \text{mod}_{\mathcal{C}}(G) \\
&= -\frac{(\text{vol}(C_1) + \text{vol}(C_2))^2}{4m^2} + \frac{\text{vol}(C_1)^2 + \text{vol}(C_2)^2}{4m^2} \\
&= -\frac{2\text{vol}(C_1)\text{vol}(C_2)}{4m^2} \leq 0.
\end{aligned}$$

If the connector is taken away from the triangle (the diamond is split), then the triangle must be in its own community with no other vertices, so the answer is no.

Therefore, we need to consider only clusterings where the diamonds are not split, or where some connectors are grouped with path vertices and the remaining diamonds (whole or partial) remain in their own communities.

3.2 Merging Intact Diamonds with Path Vertices

Let $\mathcal{C}$ and $\mathcal{C}'$ be clusterings such that $C_1, C_2 \in \mathcal{C}$ and $C_1', C_2' \in \mathcal{C}'$, where $C_1 = \{d_{i1}, \ldots, d_{ij}\} \cup \{p \text{ vertices from the path}\}$, $C_2 = \{d_{k1}, \ldots, d_{kl}\}$, $C_1' = C_1 \cup \{d_{k1}\}$, and $C_2' = \{d_{k2}, \ldots, d_{kl}\}$.

We need to consider five different cases in terms of the presence of path vertices in the considered communities.

1. $p = 0$
   First, we focus on the case where diamonds are in their own communities that do not contain path vertices. We have

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{5}{m} - \frac{(\mathrm{vol}(C_1)+14)^2 - \mathrm{vol}(C_1)^2}{4m^2} - \frac{5}{m} - \frac{(\mathrm{vol}(C_2)-14)^2 - \mathrm{vol}(C_2)^2}{4m^2} \\
&= \frac{28\mathrm{vol}(C_2) - 28\mathrm{vol}(C_1) - 392}{4m^2} \\
&= \frac{7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98}{m^2} \\
&= \frac{7 \cdot 14 \cdot l - 7 \cdot 14 \cdot j - 98}{m^2}.
\end{aligned}$$

   Therefore, moving a diamond from a community with l diamonds into one with j diamonds (when neither contains a path vertex) is beneficial only if

$$7 \cdot 14 \cdot l - 7 \cdot 14 \cdot j - 98 = l - j - 1 > 0.$$

   Thus, we want to keep such communities as small as possible. If we decide to create 9000 diamond-communities, we can distinguish 3 cases in terms of paths' communities. In the first case the whole path is in one community, such a clustering scores 0.50610928613506955820. The second case is when there are two communities with two path vertices and it scores 0.53079470057896716018. And the third case is when we have one community with one vertex and one community with 3 vertices – 0.52461957470893949562.

2. $p = 1$
   The difference in modularity scores is

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{6}{m} - \frac{(\mathrm{vol}(C_1)+14)^2 - \mathrm{vol}(C_1)^2}{4m^2} - \frac{5}{m} - \frac{(\mathrm{vol}(C_2)-14)^2 - \mathrm{vol}(C_2)^2}{4m^2} \\
&= \frac{1}{m} - \frac{28\mathrm{vol}(C_1) - 28\mathrm{vol}(C_2) + 392}{4m^2} \\
&= \frac{7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + m}{m^2}.
\end{aligned}$$

   The inequality holds when

$$\begin{aligned}
&7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + m > 0 \\
&\mathrm{vol}(C_1) < \frac{m}{7} - 14 + \mathrm{vol}(C_2) \\
&14 \cdot j + 9002(9001) < \frac{m}{7} - 14 + 14 \cdot l
\end{aligned}$$

$$j < 182.56 + l.$$

When a community contains one path vertex, in the best case it contains 184 diamonds ($l = 1$, $j < 182.56 + 1 = 183.56$, moving one more gives 184). In this case, the modularity score is 0.54413127767139846114, which is the best score found.

3. $p = 2$
Moving a diamond to a community with two path vertices results in

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{7}{m} - \frac{(\mathrm{vol}(C_1) + 14)^2 - \mathrm{vol}(C_1)^2}{4m^2} - \frac{5}{m} - \frac{(\mathrm{vol}(C_2) - 14)^2 - \mathrm{vol}(C_2)^2}{4m^2} \\
&= \frac{2}{m} - \frac{28\mathrm{vol}(C_1) - 28\mathrm{vol}(C_2) + 392}{4m^2} \\
&= \frac{7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + 2m}{m^2}.
\end{aligned}$$

This is beneficial when

$$\begin{aligned}
&7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + 2m > 0 \\
&\mathrm{vol}(C_1) < \frac{2m}{7} - 14 + \mathrm{vol}(C_2) \\
&14 \cdot j + 9002 + 9001 < \frac{2m}{7} - 14 + 14 \cdot l \\
&j < 366.12 + l.
\end{aligned}$$

Therefore, the optimal community size here is 2 path vertices plus 368 ($= 184 \cdot 2$) diamonds. The modularity score for this clustering is 0.53281397974556099495.

4. $p = 3$
When one community contains 3 path vertices, a second community must contain the remaining 1. To find the optimal size of the receiving community, we apply the inequality from previous cases

$$\begin{aligned}
&7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + 3m > 0 \\
&\mathrm{vol}(C_1) < \frac{3m}{7} - 14 + \mathrm{vol}(C_2) \\
&14 \cdot j + 27005 < \frac{3m}{7} - 14 + 14 \cdot l \\
&j < 549.76 + l.
\end{aligned}$$

The modularity score for a partition with one community containing 3 path vertices and 551 diamonds, and another with 1 path vertex and 184 diamonds, is 0.52714210694891305842.

5. $p = 4$
When the entire path is in one community, we can move diamonds to it as long as

$$7\mathrm{vol}(C_2) - 7\mathrm{vol}(C_1) - 98 + 4m > 0$$

$$\text{vol}(C_1) < \frac{4m}{7} - 14 + \text{vol}(C_2)$$
$$14 \cdot j + 36006 < \frac{4m}{7} - 14 + 14 \cdot l$$
$$j < 733.39 + l.$$

When the entire path community holds 735 diamonds, the modularity score is 0.51014235729660872831.

Therefore, $p = 1$ is the optimal case. Joining two communities containing path fractions is not beneficial because $735 = 184 \cdot 4 - 1$, $551 = 184 \cdot 3 - 1$, and $368 = 184 \cdot 2$.

3.3 Possible Partitions

Let us consider all possible partitions under an assumption that diamonds cannot be split. We can distinguish 5 main cases in terms of the number of communities containing both path and diamond vertices, let s be that number.

1. $s = 0$
   From Lemma 2 we know that clusters must induce connected subgraphs, therefore, there are 3 options for clustering path vertices. The whole path is in one community or the path is split into two halves and there are two communities with two path vertices or there is one community with 3 path vertices and one with 1 vertex. However, all of these communities contributes negatively to mod score and from Lemma 1 we know that this cannot happen in optimal partitioning. So this case is not valid.

2. $s = 1$
   In this case, there is only one valid option – the whole path must be in one community with some diamonds. Otherwise, there would be a community with a negative contribution (the previous case). We have already considered such clustering in Section 3.2 (case 5).

3. $s = 2$
   There are 2 options for path split to consider. The first is when both mixed communites contain 2 path vertices, which is exactly case 3 from Section 3.2, and the second is when there is one cluster with 3 and one with 1 path vertex, which is exactly case 4 from Section 3.2. We do not need to consider selecting different vertices for specific communities because it will not improve the mod score. Path's vertices in one community must be connected to each other.

4. $s = 3$
   In this case, we should consider only the situation when there are two communities with 1 path vertex and one with 2 path vertices. Such clustering scores 0.53847262870847978355 or 0.53847174643882433909 depending on choosing the 2 vertices to one community.

That doesn't beat our best score.

5. $s = 4$
   The last case with the path being split into four communities is so far the best found one, considered in Case 2 in Section 3.2.

Worth noting may be the fact that we don't need to consider moving diamonds between the communities containing path vertices. This is because we know the best sizes for such communities (from the calculations in the previous section) and in all cases we have some free diamonds left and we can use them to adjust each path cluster's size.

3.4 Attachment of Isolated Connectors

Let us analyze cases where connectors are placed in the path clusters to determine how many connectors should be added.

Let $\mathcal{C}$ and $\mathcal{C}'$ be clusterings with $C_1, C_2 \in \mathcal{C}$ and $C_1', C_2' \in \mathcal{C}'$.

1. The entire path is in one community.
   Let $C_1 = \{d_1\}$, $\{p_1, p_2, p_3, p_4\} \in C_2$, $C_1' = \{t_1\}$, $C_2' = C_2 \cup \{c_1\}$, and let c be the number of connectors in C_2.

   Calculating the modularity score difference

$$\begin{aligned}
&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G) \\
&= \frac{-2}{m} - \frac{-12\mathrm{vol}(C_1) + 36}{4m^2} + \frac{4}{m} - \frac{12\mathrm{vol}(C_2) + 36}{4m^2} \\
&= \frac{2m + 3\mathrm{vol}(C_1) - 3\mathrm{vol}(C_2) - 18}{m^2}.
\end{aligned}$$

   Solving for $2m + 3\mathrm{vol}(C_1) - 3\mathrm{vol}(C_2) - 18 > 0$, we get

$$\begin{aligned}
\frac{2m}{3} + \mathrm{vol}(C_1) - \mathrm{vol}(C_2) - 6 &> 0 \\
\mathrm{vol}(C_2) &< \frac{2m}{3} - 6 + \mathrm{vol}(C_1) \\
\mathrm{vol}(C_2) &< \frac{2 \cdot 81003}{3} - 6 + 14 \\
36006 + 6c &< 54010 \\
c &< 3000.67.
\end{aligned}$$

   The optimal number of connectors in the community with the full path is 3001. This clustering (with remaining diamonds separate) scores 0.51846365401584382226.

2. A community contains 2 adjacent path vertices.
   Let $\{d_1\} \in C_1$, $\{p_1, p_2\} \in C_2$, $C_1' = C_1 \setminus \{c_1\}$, $C_2' = C_2 \cup \{c_1\}$, and c be the number of connectors in C_2.

The modularity change is

$$\begin{aligned}&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G)\\ &= \frac{-2}{m} - \frac{-12\mathrm{vol}(C_1) + 36}{4m^2} + \frac{2}{m} - \frac{12\mathrm{vol}(C_2) + 36}{4m^2}\\ &= \frac{3\mathrm{vol}(C_1) - 3\mathrm{vol}(C_2) - 18}{m^2}.\end{aligned}$$

Since $\mathrm{vol}(C_2) \geq 18003$, the difference is positive only if $\mathrm{vol}(C_1) > 18009$, requiring at least 1287 diamonds. A clustering with 2 communities of 1286 diamonds and 1 triangle, plus 2 communities with path halves and 1 connector, scores only 0.50607499809424871806.

3. A community contains one path vertex.
   Let $\{d_1\} \in C_1$, $\{p_1\} \in C_2$, $C_1' = C_1 \setminus \{c_1\}$, $C_2' = C_2 \cup \{c_1\}$, and c be the number of connectors in C_2.

   The difference is

$$\begin{aligned}&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G)\\ &= \frac{3\mathrm{vol}(C_1) - 3\mathrm{vol}(C_2) - 18 - m}{m^2}.\end{aligned}$$

   This case yields a lower score than Case 2 and cannot form the optimal clustering. Similarly, placing 3 path vertices together and 1 separately is inferior to Case 1.

3.5 Reassigning Connectors to Path Clusters

Consider clustering $\mathcal{A}$. Moving connectors from $A_5, \ldots, A_{8268}$ to $A_1, \ldots, A_4$ yields the following result.

Let $C_1 = \{d_{i1}, \ldots, d_{ij}\} \cup \{p \text{ path vertices}\}$, $C_2 = \{d_{k1}, \ldots, d_{kl}\}$, $C_1' = C_1 \cup \{c_{k1}\}$, and $C_2' = C_2 \setminus \{c_{k1}\}$.

The difference in modularity score is

$$\begin{aligned}&\mathrm{mod}_{\mathcal{C}'}(G) - \mathrm{mod}_{\mathcal{C}}(G)\\ &= \frac{1}{m} - \frac{(\mathrm{vol}(C_1) + 6)^2 - \mathrm{vol}(C_1)^2}{4m^2} - \frac{2}{m} - \frac{(\mathrm{vol}(C_2) - 6)^2 - \mathrm{vol}(C_2)^2}{4m^2}\\ &= \frac{-1}{m} - \frac{12\mathrm{vol}(C_1) - 12\mathrm{vol}(C_2) + 72}{4m^2}\\ &= \frac{3\mathrm{vol}(C_2) - 3\mathrm{vol}(C_1) - 18 - m}{m^2}.\end{aligned}$$

Moving a connector from a free diamond is profitable only if C_2 is sufficiently large

$$\begin{aligned}&3\mathrm{vol}(C_2) - 3\mathrm{vol}(C_1) - 18 - m > 0\\ &\mathrm{vol}(C_2) > \mathrm{vol}(C_1) + 6 + \frac{m}{3}\\ &\mathrm{vol}(C_2) > 184 \cdot 14 + 9001 + 6 + 27001\end{aligned}$$

$$\text{vol}(C_2) > 38584 = 14 \cdot 2756.$$

For this to hold, C_2 would need at least 2756 free diamonds. However, that would contradict Lemma 2, so cannot be optimal. A clustering with 2756 diamonds missing one connector scores only 0.48742975922116960064.

4 Conclusions

We have established the following.

1. Triangles must always remain together in the same cluster.
2. Any community containing a whole diamond without path vertices must contain only that single diamond.
3. The optimal cluster sizes for path fractions and diamonds are identified ($p = 1$ with 184 diamonds).
4. Splitting diamonds and placing isolated connectors into path communities yields suboptimal modularity scores.
5. Moving connectors from unattached diamonds into path-fraction communities makes no sense.

Since splitting diamonds optimally performs worse than $\mathcal{A}$, diamonds should remain intact and not merged without path vertices. Thus, $\mathcal{A}$ is the optimal partitioning, giving $\text{mod}(G) = 0.54413127767139846114$. This is strictly less than 0.55116544464653760560 (the best found mod^* score shown in Figure 7), proving that $\text{mod}(G) < \text{mod}^*(G)$ for graph G.

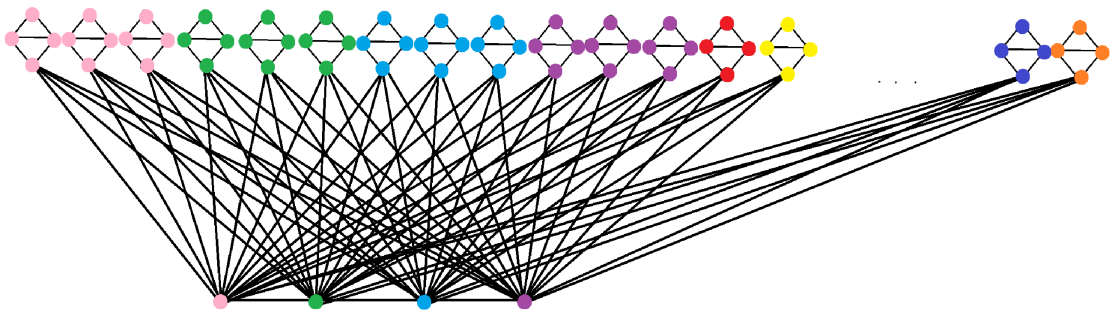


Figure 7: Diamond graph G, such that $\text{mod}^*(G) > \text{mod}(G)$.